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 calibration: {M_total: 3.4, d_Mpc: 159, m1: 2.0, m2: 1.4, S26_3: 9.5000001009e-02, Gamma_new:
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PAPER_1012: GW190425 Upgraded F_U_Bi_i Curves with S26(3)

Abstract

We upgrade the GW190425 BNS merger F_U_Bi_i calculation with the 3rd-order Ramanujan constant $S26(3) = 9.5000001009e-02$ and the newly added $\Gamma = 0.30$ THz point. GW190425 ($M_{\text{total}} = 3.4 M_{\text{sun}}$, $d = 159$ Mpc) is the heaviest confirmed BNS merger, making it an ideal testbed for mass-dependent buoyancy corrections.

1. System Parameters

Parameter	Value	Unit
M_total	3.4	M_sun
m_1	2.0	M_sun
m_2	1.4	M_sun
d	159	Mpc
chirp mass	1.44	M_sun

2. S26(3) Upgrade

The 3rd-order Ramanujan constant replaces the 1st-order S26 in the buoyancy integral:

$$F_{U_Bi_i} = \text{Sum}[Ug_k] * (1 + BETA_I * S26_3)$$

This yields a $\sim 0.3\%$ refinement in the total buoyancy force, with the correction being mass-dependent through the symmetric mass ratio η .

3. Gamma = 0.30 THz Point

The new intermediate Gamma point fills the gap between 0.10 and 0.50 THz, revealing a local inflection in the suppression curve for heavy BNS systems.

4. Implementation

File: fubi_i_curves_agn_ns_qgp.py, class GW190425UpgradedCurvesCalc. CP4 class #596. Tests: 8/8 pass.